Frontend Architecture Overview

The frontend is built using **Next.js 15.4.3** with **React 19.1.0**, providing a modern, performant web application with server-side rendering capabilities. Here's a comprehensive breakdown:

🛠️ Technology Stack

Core Dependencies:

- **Next.js 15.4.3** - React framework with SSR/SSG
- **React 19.1.0** - Latest React with concurrent features
- **TypeScript 5.x** - Type-safe JavaScript
- **Tailwind CSS 4.x** - Utility-first CSS framework
- **Radix UI** - Accessible component primitives
- **Recharts** - React charting library
- **React Hook Form** - Form state management
- **Zod** - Schema validation
- **Axios** - HTTP client for API calls

Development Tools:

- **Jest** - Testing framework
- **Testing Library** - React component testing
- **ESLint** - Code linting
- **Tailwind CSS** - Modern CSS framework

🎨 Design System & Styling

Tailwind CSS Configuration:

- **Custom CSS Variables**: Comprehensive design token system
- **Dark Mode Support**: Built-in dark/light theme switching
- **Responsive Design**: Mobile-first approach
- **Custom Animations**: Smooth transitions and loading states

UI Components (`/components/ui/`):

```
ui/
├── button.tsx         # Reusable button components
├── card.tsx           # Card layout components
├── input.tsx          # Form input components
├── label.tsx          # Form label components
├── alert.tsx          # Alert/notification components
├── navigation-menu.tsx # Navigation components
└── sonner.tsx         # Toast notification system
```

Features:

- **Accessible**: Built on Radix UI primitives
- **Customizable**: Class variance authority for variants
- **Type-safe**: Full TypeScript support
- **Consistent**: Unified design language

📱 Application Structure

App Router Structure (`/src/app/`):

```
app/
├── layout.tsx         # Root layout with navigation
├── page.tsx           # Home page
├── globals.css        # Global styles
├── log/
```

```
│   └── page.tsx       # Water logging page
└── summary/
    └── page.tsx       # Analytics summary page
```

Component Architecture:

```
components/
├── Navigation.tsx          # Main navigation bar
├── WaterLogForm.tsx        # Water intake form
├── WaterSummaryChart.tsx   # Analytics chart
├── ErrorBoundary.tsx       # Error handling
├── Loading.tsx             # Loading states
└── ui/                     # Reusable UI components
```

🌐 Page Structure & Routing

1. Home Page (`/`) - Landing Page

```
// Features:
- Welcome message and app description
- Quick action cards for Log and Summary
- Daily hydration goal display (2000ml)
- Responsive grid layout
- Hover effects and transitions
```

2. Log Page (`/log`) - Water Intake Form

```
// Features:
- Water intake logging form
- Date and amount input fields
- Quick preset buttons (250ml glass, 500ml bottle)
- Real-time validation
- Success/error feedback
- Hydration tips section
```

3. Summary Page (`/summary`) - Analytics Dashboard

```
// Features:
- 7-day water intake chart
- Statistics cards (average, goal achievement, daily target)
- Interactive bar chart with tooltips
- Goal reference line
- Loading and error states
```

🧭 Navigation System

Navigation Component:

```
// Features:
- Responsive navigation bar
- Active page highlighting
- Icon-based navigation
- Mobile-friendly design
- Brand identity with water droplet icon
```

Navigation Items:

- **Log Water** (`/log`) - Water intake form
- **Summary** (`/summary`) - Analytics dashboard

📊 Data Management & API Integration

API Layer (`/lib/api.ts`):

```
// Configuration:
- Axios client with base URL configuration
- Environment variable support (NEXT_PUBLIC_API_URL)
- Type-safe API interfaces
- Error handling

// Endpoints:
- POST /water-log - Log water intake
- GET /water-summary/:userId - Get 7-day summary
```

Custom Hooks (`/hooks/useWaterApi.ts`):

`useWaterLog()` Hook:

```
// Features:
- Form submission state management
- Loading, error, and success states
- Automatic state reset
- Error handling with user-friendly messages
```

`useWaterSummary()` Hook:

```
// Features:
- Data fetching with loading states
- Error handling
- Automatic data refresh
- Memoized fetch function
```

📝 Form Handling & Validation

WaterLogForm Component:

```
// Features:
- React Hook Form integration
- Zod schema validation
- Real-time error display
- Quick preset buttons
- Form reset functionality
- Loading states during submission
```

Validation Schema (`/lib/validations.ts`):

```
export const waterLogSchema = z.object({
  date: z.string().min(1, 'Date is required'),
  intakeMl: z
    .number()
```

```
    .min(1, 'Intake must be at least 1ml')
    .max(10000, 'Intake cannot exceed 10,000ml'),
});
```

Validation Rules:

- **Date**: Required, must be valid date string
- **Intake**: Number between 1-10,000ml
- **Real-time**: Instant feedback on form errors

📈 Data Visualization

WaterSummaryChart Component:

```
// Features:
- Recharts integration for responsive charts
- Bar chart with goal reference line
- Interactive tooltips
- Statistics cards with key metrics
- Loading and error states
- Responsive design
```

Chart Features:

- **7-day view**: Shows current week's data
- **Goal line**: Red dashed line at 2000ml
- **Tooltips**: Detailed information on hover
- **Statistics**: Average intake, goal achievement rate
- **Responsive**: Adapts to different screen sizes

🎯 User Experience Features

1. Quick Actions:

- **Preset buttons**: Glass (250ml) and Bottle (500ml)
- **One-click logging**: Streamlined workflow
- **Form reset**: Easy form clearing

2. Visual Feedback:

- **Loading states**: Spinner animations
- **Success messages**: Green alerts with checkmarks
- **Error handling**: Red alerts with error icons
- **Hover effects**: Interactive elements

3. Responsive Design:

- **Mobile-first**: Optimized for all screen sizes
- **Touch-friendly**: Large touch targets
- **Readable typography**: Clear text hierarchy

4. Accessibility:

- **Semantic HTML**: Proper heading structure
- **ARIA labels**: Screen reader support
- **Keyboard navigation**: Full keyboard support
- **Color contrast**: WCAG compliant colors

🧪 Testing Setup

Jest Configuration:

```
// Features:
- Next.js Jest integration
- Testing Library setup
- Module path mapping (@/ alias)
- JSDOM environment
- Coverage reporting
```

Test Structure:

```
__tests__/
├── components/
│   └── WaterLogForm.test.tsx
└── hooks/
    └── useWaterApi.test.tsx
```

🔧 Utility Functions (`/lib/utils.ts`)

Key Utilities:

```
// Date formatting
- formatDate(): Converts Date to YYYY-MM-DD
- formatDateDisplay(): Formats for chart display

// Styling
- cn(): Class name merging with Tailwind

// Demo utilities
- getDefaultUserId(): Demo user ID for testing
```

🎨 Styling & Theming

Global Styles (`globals.css`):

```
// Features:
- Tailwind CSS imports
- Custom CSS variables
- Dark mode support
```

113. - Custom animations
114. - Focus states
115. - Loading animations

Design Tokens:

- **Colors**: Primary blue (#3b82f6), success green, error red
- **Typography**: Inter font family
- **Spacing**: Consistent spacing scale
- **Border radius**: Rounded corners for modern look

📱 Responsive Design

Breakpoint Strategy:

- **Mobile**: < 768px (default)
- **Tablet**: 768px - 1024px
- **Desktop**: > 1024px

Responsive Features:

- **Flexible grids**: CSS Grid with responsive columns
- **Adaptive navigation**: Collapsible mobile menu
- **Scalable charts**: Responsive chart containers
- **Touch-friendly**: Large buttons and inputs

🔒 Security & Best Practices

Security Features:

- **Input validation**: Client and server-side validation
- **XSS protection**: React's built-in XSS protection
- **CORS handling**: Proper API communication
- **Type safety**: Full TypeScript coverage

Performance Optimizations:

- **Code splitting**: Automatic Next.js code splitting
- **Image optimization**: Next.js image optimization
- **Bundle analysis**: Built-in bundle analyzer
- **Lazy loading**: Component lazy loading

📁 File Structure Summary

```
frontend/
├── src/
│   ├── app/                  # Next.js App Router
│   │   ├── layout.tsx        # Root layout
│   │   ├── page.tsx          # Home page
│   │   ├── log/page.tsx      # Logging page
│   │   ├── summary/page.tsx  # Analytics page
│   │   └── globals.css       # Global styles
│   ├── components/           # React components
│   │   ├── Navigation.tsx    # Main navigation
│   │   ├── WaterLogForm.tsx  # Water intake form
│   │   ├── WaterSummaryChart.tsx # Analytics chart
│   │   └── ui/               # Reusable UI components
│   ├── hooks/                # Custom React hooks
│   │   └── useWaterApi.ts    # API integration hooks
│   └── lib/                  # Utility libraries
│       ├── api.ts            # API client
│       ├── utils.ts          # Utility functions
│       └── validations.ts    # Form validation schemas
├── public/                   # Static assets
├── package.json              # Dependencies & scripts
└── jest.config.js            # Testing configuration
```

🎯 Key Features Summary

1. **Modern Stack**: Next.js 15 + React 19 + TypeScript
2. **Beautiful UI**: Tailwind CSS + Radix UI components
3. **Type Safety**: Full TypeScript implementation
4. **Form Handling**: React Hook Form + Zod validation
5. **Data Visualization**: Interactive charts with Recharts
6. **Responsive Design**: Mobile-first approach
7. **Testing**: Jest + Testing Library setup
8. **Accessibility**: WCAG compliant design
9. **Performance**: Optimized with Next.js features
10. **Developer Experience**: Hot reload, linting, formatting